Here is the complete **Python Flask script** to process incoming text messages from WhatsApp.

When someone sends a message to your WhatsApp Business number, Meta sends a POST request to your permanent ngrok URL. This code automatically navigates through Meta's deeply nested JSON structure to extract the **sender's phone number**, **sender's name**, and the **text message** they sent.

The Python Webhook Script

```
from flask import Flask, request, jsonify

app = Flask(__name__)

# Must match the token you put inside the Meta Developer Portal
WEBHOOK_VERIFY_TOKEN = "your_chosen_secret_verify_token"

@app.route('/webhook', methods=['GET', 'POST'])
def webhook():
    # 1. HANDLE META VERIFICATION (GET Request)
    if request.method == 'GET':
        mode = request.args.get('hub.mode')
        token = request.args.get('hub.verify_token')
        challenge = request.args.get('hub.challenge')

        if mode == 'subscribe' and token == WEBHOOK_VERIFY_TOKEN:
            print("✅ Webhook verified successfully by Meta!")
            return challenge, 200
        else:
            print("❌ Verification failed: Token mismatch.")
            return "Forbidden", 403

    # 2. HANDLE INCOMING MESSAGES (POST Request)
    elif request.method == 'POST':
        data = request.json

        # Safe checking to ensure it's a valid WhatsApp message
payload
        if data.get("object") == "whatsapp_business_account":
            try:
                # Drill down into Meta's nested payload structure
                entry = data["entry"][0]
                changes = entry["changes"][0]
                value = changes["value"]
```

```
                # Check if there are actual messages in the payload
                if "messages" in value:
                    message_details = value["messages"][0]
                    contact_details = value["contacts"][0]

                    # Extract the useful information
                    sender_phone = message_details["from"]
# e.g., "15550123456"
                    sender_name =
contact_details["profile"]["name"] # e.g., "John Doe"

                    # Check if the message is a text message
                    if message_details["type"] == "text":
                        message_text =
message_details["text"]["body"]

                        # PRINT THE MESSAGE TO YOUR TERMINAL
                        print("\n--- New WhatsApp Message ---")
                        print(f"From: {sender_name}
({sender_phone})")
                        print(f"Message: {message_text}")
                        print("----------------------------\n")

                        # TODO: Add your logic here (e.g., save to
DB or trigger a reply)

            except (KeyError, IndexError) as e:
                # Ignore status updates or read-receipt deliveries
sent by Meta
                pass

        # Always return 200 OK to Meta so they know you received
the event
        return jsonify({"status": "success"}), 200

if __name__ == '__main__':
    # Start the local server on port 5000
    app.run(port=5000, debug=True)
```

Steps to Test it Right Now:

1. **Run your Python script** in your cloud/local environment.
2. **Start ngrok** using your permanent free domain:
   `ngrok http --url=yourname-xyz.ngrok-free.app 5000`

3. Go to the **Meta Developer Portal**, send a test message from the "WhatsApp Getting Started" panel, or text your test phone number directly from your phone.

4. Watch your **Python terminal**—the message text and sender name will print directly onto your screen in real time.

Once you see the messages coming in, do you want to see how to **send an automated reply back to the user** using a Python requests call to Meta's Cloud API?